Types of data

1 State whether the following data are numerical or categorical:

a Amount owing on layby.

b Types of vegetables.

c Brands of tablets.

d Types of folders.

e Colours of folders.

f Brands of phones.

g Maximum snowfall.

h Daily UV index.

i Types of dogs.

j Favourite colours.

k Cost of a PS4 game.

l Types of fruits.

2 State whether the following examples of numerical data are discrete or continuous:

a The number of classrooms in your school.

b Daily humidity.

c The ages of a group of people.

d The time taken to run 200 metres.

e Amount of money paid for walking the dog.

f The number of people taller than you in your class.

g The top speed of different models of motorcycles.

h The size of people's rare coin collections.

i Lengths of cats' whiskers.

j The distance from the Earth to nearby galaxies.

k How long it takes you to fall asleep at night.

l The amount of soft drink you drink in a day.

m The number of cans of soft drink you drink in a day.

n The world record time for solving ten Rubik's cubes.

o The hourly pay rate for a particular job.

p The rainfall in your city.

q Number of children in your sports team.

r Amount of money owing on a loan.

s Your marks in your most recent Science test.

t The number of views that a YouTube channel has.

u The strength of an earthquake

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3 For each of the following survey questions, classify the answer as categorical, discrete numerical or continuous numerical data:

- **a** How far do you travel to school each day?
- **b** Where was your most recent holiday?
- **c** How many siblings do you have?
- **d** What modes of transport do you use to get to school?
- **e** How long did you sleep for last night?
- **f** How many people live in your house?
- **g** How long does it take for you to get to school everyday?
- **h** What languages do you speak at home?
- **i** What kinds of pets do you own?
- **j** How many pets do you own?
- **k** How many times have you broken your arm?
- **l** How long did you spend on homework last night?
- **m** How tall are you?
- **n** How much does your schoolbag weigh?
- **o** How many apps do you have in your phone?
- **p** Do you think recess is long enough?

Tables

4 Consider the given data set that shows the distance to the nearest hospital (in km):

$$26.4, \quad 28.7, \quad 15.1, \quad 37.5, \quad 11.8, \quad 7.9, \quad 16.7, \quad 6.9, \quad 5, \quad 24.9, \quad 35.3, \quad 45.5$$

- **a** Is the data type continuous or discrete?
- **b** What is the farthest distance to the nearest hospital?

5 Glen surveys his class about their favourite movie genre. His results are shown in the table:

Genre	Drama	Comedy	Action	Horror
Students	5	7	6	2

- **a** What type of data is displayed in the table?
- **b** Which genre is the most popular among his class?
- **c** How many people did Glen survey?

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6 Ten people were asked about the number of hours they spent outside last weekend. The results of the survey are shown:

$$5, \quad 9, \quad 9, \quad 4, \quad 6, \quad 9, \quad 4, \quad 9, \quad 6, \quad 4$$

a What type of data is this?

b How many people spent exactly 6 hours outside?

c What was the longest that anyone spent outside?

d What was the most common amount of time spent outside?

7 Ten people were asked about the number and type of pets they own. The results of the survey are shown below:

Twelve fish	Three turtles	Two cats	Two cats	One dog
One dog	One dog, One cat	One cat	Five ferrets	One dog

a What type of data is the type of pet?

b How many cats do these ten people own altogether?

c How many pets do these ten people own altogether?

d What is the most common survey response?

Question biases

8 State whether the following questions are biased or fair?

a Do you want a soup for lunch or a sandwich?

b What do you do on a Sunday morning?

c Do you like these shoes?

d Do you prefer this shirt or the one on the shelves at the moment?

e Do you think the government should be allowed to cut down some of the oldest trees in the area to construct a metro railway line in the city?

f Do you prefer newspapers or news on television?

g Do you prefer the full time degree program or part time degree program?

h Should the government enforce a minimum drinking age for its citizens?

i Do you eat at least the recommended number of servings of fruits and vegetables to ensure a healthy and long life?

j Do you think bike helmets should be mandatory for all bike riders?

k Do you prefer the natural beauty of hardwood floors in your home?

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l Do you exercise regularly?

m Do you feel that the TV news is an inaccurate portrayal of life's problems?

n Don't you think this newspaper is biased?

o Do you prefer the look and feel of thick lush carpeting in your living room?

p Do you take these extra strength multi-vitamins to supplement your diet?

9 Explain why the following survey questions are poor:

a The Prime Minister believes that taxes are too high, do you think taxes are too high?

b Do you have a Snapchat or Twitter account?

c Do your siblings ever drive you to school?

d Do you often waste time on social media?

e What brand of shoe polish do you use everyday?

10 Patricia surveys her class about their favourite music. Her results are shown in the table:

Genre	Country	Pop
No. Students	15	2

a According to the survey, which genre is most popular among her class?

b This was her survey question:

"The coolest kids like Country music, and nobody likes Pop. Do you like Country music or Pop music?"

Do you trust the results of Patricia's survey? Explain your answer.

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